

Week 2 - Day 1 - Monday - Installing R; Getting QSS to work; and some R basics

2024-01-05

For those of you who haven't yet - here is where you can install R and Rstudio:

<https://posit.co/download/rstudio-desktop/>

Make sure to install R first, and also choose the version for your operating system.

As you heard in lecture, you will need to submit several tutorials from the QSS book. These tutorials can be accessed through R.

To do so, you will first need to install some packages:

```
# remotes::install_github("kosukeimai/qss-package", build_vignettes = TRUE)
# remotes::install_github("rstudio/learnr")
# remotes::install_github("rstudio-education/gradethis")
# remotes::install_github("mattblackwell/qsslearnr")
```

If you are experiencing issues, please visit this website first: <https://github.com/mattblackwell/qsslearnr>

To run the tutorial, execute the following code:

```
# learnr::run_tutorial(package = "qsslearnr")

# learnr::run_tutorial("00-intro", package = "qsslearnr")
```

At the end of the tutorial, you will be prompted to download your report

You can then add your name and upload them on Canvas

R Basics

```
# Concatenate vectors into a new vector
c(1, 2, 3)
```

```
## [1] 1 2 3
```

```
# Assign them to a new object for manipulation
x <- c(1, 2, 3)

x <- x * 10

x
```

```
## [1] 10 20 30
```

*New assignment will overwrite the original values if you assign some values to an existing object.
This is a major source of errors. Keep distinct object names*

```
result <- 7  
result
```

```
## [1] 7
```

```
result <- 8  
result
```

```
## [1] 8
```

Remove objects with rm()

```
rm(x)  
rm(result, x)
```

```
## Warning in rm(result, x): object 'x' not found
```

Functions

```
x <- c(2, 3, 81, 58, 274)  
  
# Functions take input(s) and produce output(s)  
class(x)
```

```
## [1] "numeric"
```

```
class("Nicolas")
```

```
## [1] "character"
```

```
length(x)
```

```
## [1] 5
```

```
mean(x)
```

```
## [1] 83.6
```

The general idea is pretty simple - you use functions as a computation tool

You can create your own functions - this can get as complicated as you would like

Moving on

A good practice is to keep your projects and files organized and tidy. Avoid accumulating data and R files in your download folder. I recommend creating a folder for each week of the lab

Setting the WD

Every time you start a new project, you need to set a new directory in R, so R knows where to access files and save output.

```
#use this command to find out what your current WD is
getwd()

#use setwd() to set your working directory
#setwd("")
```

You can also save our R file in the same folder that your data is in

Or you can manually set the working directory, by clicking on Session - Set Working Directory

RMarkdown

RMarkdown files allow for easy creation of reproducible reports and presentations, with the ability to interweave code and its output, as well as integrate images and other media.

If you are looking for an introduction for RMarkdown: Chapter 27 from Wickham and Grolemund (2017) R for Data Science.

If you want a more comprehensive guide, then check out Xie et al. (2021) - **R Markdown: The Definitive Guide

Maybe most important in RMarkdown is the keyboard shortcut: Ctrl + Alt + I to create a new code chunk

In MAC, this works like this: Ctrl + Option + I

Let's try our hands at some practical applications

Let's all make sure we have the turnout.csv file downloaded from Canvas

Let's create a new folder on our desktops - or wherever

Set Working directory to the folder I want to work from

Import dataset

```
data <- read.csv("turnout.csv")
```

```
head(data)
```

##	year	VEP	VAP	total	ANES	felons	noncit	overseas	osvoters
## 1	1980	159635	164445	86515	71	802	5756	1803	NA
## 2	1982	160467	166028	67616	60	960	6641	1982	NA
## 3	1984	167702	173995	92653	74	1165	7482	2361	NA
## 4	1986	170396	177922	64991	53	1367	8362	2216	NA
## 5	1988	173579	181955	91595	70	1594	9280	2257	NA
## 6	1990	176629	186159	67859	47	1901	10239	2659	NA

```
max(data$year)
```

```
## [1] 2008
```

```
mean(data$overseas)
```

```
## [1] 2745.714
```

Dplyr Functions

```
library(dplyr)
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

Filter

```
data <- filter(data, year > 1992)
```